

Milton Tinoco

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Objective

Computer Engineering graduate with hands-on experience developing real-time embedded systems, sensor-based applications, robotics software, Linux-based platforms, computer vision systems, and machine learning models for real-world applications.

Education

San Francisco State University
M.S. Electrical and Computer Engineering
B.S. Computer Engineering

San Francisco, CA
2025–2027
2025

Santa Rosa Junior College
A.S. Mathematics; A.S. Natural Science

Santa Rosa, CA
2023

Technical Skills

Programming Languages: C/C++, Python, ARM Assembly, Bash/Linux command line

Computer Vision & ML: OpenCV, stereo vision, image preprocessing, camera calibration, stereo rectification, disparity/depth estimation, point-cloud reconstruction, YOLO, machine learning

Embedded Systems & Sensors: STM32/ARM Cortex-M, FreeRTOS, bare-metal programming, ADC, UART, GPIO, I2C, timers, DMA, ToF sensors, EMG sensors, multi-sensor integration

Robotics & Platforms: ROS/ROS2, SLAM, Nav2, LiDAR, Intel RealSense, NVIDIA Jetson, Raspberry Pi, Docker, Linux-based development

Tools & Debugging: STM32CubeIDE, Git/GitHub, Analog Discovery 2 logic analyzer, oscilloscopes, data visualization

Projects

Stereo Vision System for 3D Room Modeling | Python, OpenCV, Linux 2024–2025

- Led the development of a stereo vision system for environmental mapping and blueprint generation using Python and computer vision techniques on a Linux-based platform.
- Developed a custom computer vision pipeline with image preprocessing techniques to reduce noise, improve feature quality, and prepare visual data for accurate stereo vision and 3D reconstruction.
- Implemented camera calibration, stereo rectification, disparity mapping, and depth estimation to support 3D room modeling and environmental reconstruction.
- Developed low-level motor control drivers and interfaced with Linux system commands for hardware control, testing, and debugging using shell scripting.
- Applied deep learning-based methods to improve depth estimation accuracy and computational efficiency.
- Developed Python-based wireless communication modules using Wi-Fi and Bluetooth sockets to enable structured data exchange between the embedded device and host computer.

On-Device Waste Object Detection System | Raspberry Pi, Python, Object Detection 2025

- Developed an on-device object detection system on a Raspberry Pi to recognize waste materials and support an interactive waste classification system.
- Trained and evaluated object detection models using online datasets and custom image datasets to improve waste classification performance in real-world conditions.

ROS2 Social Guidance Robot | ROS2, SLAM, Nav2, NVIDIA Jetson 2026

- Developed an autonomous guidance robot by integrating ROS2, SLAM, Nav2, LiDAR, Intel RealSense sensors, and computer vision tools on an NVIDIA Jetson platform to assist students with room navigation inside a building.

Assistive Haptic Hat | STM32, ToF Sensors, Haptics 2026

- Developed a 360-degree obstacle detection system using ToF sensors, haptic feedback motors, gyroscopic sensing, and an STM32 microcontroller to support navigation for visually impaired individuals.

Work Experience

Graduate Researcher
San Francisco State University

May 2024–Present
San Francisco, CA

- Lead a research team developing an embedded EMG classification system for a prosthetic exoskeleton, with the current system achieving approximately 90% intent recognition accuracy.
- Manage task assignments, coordinate project goals with faculty advisors, review student-developed software, verify experimental results, prepare weekly progress reports, and maintain documentation and version control on GitHub.
- Designed embedded C/C++ firmware for an ARM Cortex-M microcontroller, integrating a machine learning classifier and custom serial communication protocol for real-time data exchange.
- Developed low-level drivers for ADC, UART, GPIO, and timers using DMA and interrupt-driven processing; performed signal logging, AD2 logic analysis, and STM32 register inspection to improve reliability.